

in Chile, Brazil, and Greece. The fact that such improvised forces performed with considerable success in the independence wars was powerful evidence of the decline of those once-great naval powers, Spain and the Ottoman Empire. The weakness of Ottoman sea power had been exposed by a series of defeats at the hands of Russia in the late 18th century. This process of humiliation was completed at the Battle of Navarino in 1827.

ON THE BRINK OF CHANGE

Navarino was still a battle from the age of sail. But the transition to a new era of naval warfare was already under way. The introduction of steam ships into warfare was slow because their usefulness was at first limited. In 1814 the United States built the first steam-powered warship, *Demologos*, designed by the prolific inventor Robert Fulton, but it was only suitable for use as a floating gun battery in harbor. The British adopted steamships as tugs, initially only to tow sailing ships in and out of harbor, but later as armed support vessels for use in combat. By 1829 the Royal Navy had built eight armed paddle steamers. The *Karteria*, built and manned by the British, saw action with the Greeks in their independence war.

Meanwhile, in 1824, French general Henri-Joseph Paixhans showed the effectiveness of a naval gun firing explosive shells. Between them, steam power and the exploding shell spelled the end of battle between wooden sailing ships. The latter would soon be replaced by ironclads.

Bombardment of Algiers
Dutch and British ships bombard Algiers in 1816, in an operation that broke the power of the Barbary pirates.